WEIGHTED DIJOIN PACKING IN ONE-EDGE DELETIONS OF CHORDAL DIGRAPHS AND SIX-TERMINAL SINGLE-DEFECT DIGRAPHS

ARGUS OPEN RESEARCH PROJECT

Abstract. We establish two unconditional extensions of Woodall's 1976 dijoin packing conjecture $\nu(D, w) = \tau(D, w)$ for nonnegative integer arc capacities $w\colon A \to \mathbb{Z}_{\geq 0}$. First, extending the 2025 theorem of Cornuéjols, Liu, and Ravi for weighted chordal digraphs, we prove that $\nu(D, w) = \tau(D, w)$ holds for every finite loopless digraph D whose underlying simple undirected graph becomes chordal after adding at most one edge. Second, in the single source-sink reachability defect setting (where the condensation has at most one unreachable ordered source-sink pair), we prove weighted dijoin packing for all topologies with at most six terminals and establish that any globally minimal missing-color configuration in an arbitrary-size single-defect digraph must have Boolean quotient rank at least five (surpassing the rank-four boundary of Williams's 2004 conjecture).

1. Main Mathematical Results

Theorem 1.1 (Weighted dijoin packing for one-fill chordal and six-terminal single-defect digraphs)**.** (1) *For every finite loopless digraph $D = (V, A)$ whose underlying simple graph is chordal or becomes chordal upon adding one edge, and every $w\colon A \to \mathbb{Z}_{\geq 0}$ with $\tau(D, w) < \infty$, $\nu(D, w) = \tau(D, w)$.*

(2) *For every finite directed multigraph D with $w\colon A \to \mathbb{Z}_{\geq 0}$ whose condensation has at most one unreachable source-sink pair and at most six terminal vertices, $\nu(D, w) = \tau(D, w)$; furthermore, any minimal missing-color configuration in a general single-defect digraph has Boolean quotient rank at least* 5.

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 3256 in the domain of combinatorial optimization and directed graphs adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Weighted dijoin packing nu(D,w)=tau(D,w) for all one-fill chordal digraphs and all 6-terminal one-defect digraphs, plus quotient-rank >= 5 lower bound on minimal missing-color configurations.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Unrestricted Woodall's conjecture for general digraphs

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 3256 (combinatorial optimization and directed graphs), archived in the accompanying repository package `results/3256/`.

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